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Sardar Patel University

SYBCA -4th Sem (Reg & NC) External Examination-AUG-2022

US04CBCA27 || Operating System

Date: 24/8/2022

Total Marks: 70

Time: 10:00 am to 12:00 pm

Q. 1 Multiple Choice Questions

[10]

- 1 _____ Scheduler select process from ready queue for execution.
(a) Short-term (b) Long-term (c) Medium-term (d) None of these
- 2 OS table is also called _____.
(a) Job table (b) Process table (c) Ready queue (d) Buffer
- 3 Multi-tasking system also known as _____.
(a) Real-time (b) Multi-User (c) Time-Sharing (d) None of these
- 4 Which of the following memory allocation strategy is fastest?
(a) Best-fit (b) Worst-fit (c) Optimal-fit (d) First-fit
- 5 In Paging, Physical memory is divided into fixed-size blocks is called
(a) Frames (b) Pages (c) Files (d) None of these
- 6 A Lazy Swapper is also known as _____.
(a) Frames (b) Pager (c) Pages (d) None of these
- 7 _____ buffer has fixed buffer size.
(a) Shared (b) Unbounded (c) Bounded (d) None
- 8 The concept in which output depends on sequence of process execution is known as _____.
(a) Output-Condition (b) Paging (c) Deadlock (d) Race-Condition
- 9 Which command is used to list all the files in your current directories (including hidden)?
(a) ls -a (b) ls -d (c) ls -b (d) ls -c
- 10 For getting present working directory, _____ command is used.
(a) wd (b) pwd (c) pd (d) dir

Q. 2 Fill in the blanks and True/False.

[08]

- 1 Autopilot in an aircraft is an example of _____ type of OS.
- 2 Valid-Invalid bit scheme in page table is used in _____
- 3 _____ buffer has fixed buffer size.
- 4 _____ command is used to get information about currently logged in user on to system.
- 5 A Program in execution is known as Files. (True / False).
- 6 FIFO page replacement algorithm suffers from Belady's anomaly. (True / False).
- 7 The process which cannot affect or be affected by the other processes executing in the system is known as Cooperative process. (True / False).
- 8 cd command is used to change the working directory. (True / False).

Q. 3 Answer the following. (Attempt any 10)

[20]

- 1 Draw the diagram of PCB.
- 2 List out services of OS.
- 3 Explain Key elements of OS.
- 4 Explain Internal Fragmentation in short.
- 5 What is Compaction? What is the use of it?
- 6 Calculate Page faults using FIFO algorithm for following reference string:
(Number of Frames = 3)
Reference string = 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,
- 7 Explain any two necessary conditions of Deadlock.
- 8 Draw the general structure of process.
- 9 Explain algorithm-1 for two-process solution in brief.
- 10 Discuss mkdir and mv in brief.
- 11 Explain wc in brief.
- 12 Write use of echo command.

Q. 4 Answer the following (Attempt any 4)

[32]

- 1 Explain in details process state and its transitions.
- 2 Explain SJF scheduling algorithm with example.
- 3 Explain FIFO page replacement algorithm with advantages and disadvantages
- 4 What is Memory Management? Explain Memory Management Allocation Techniques
- 5 What is Co-operative Process? Explain Producer-Consumer Problem.
- 6 What is LINUX? Explain basic features of LINUX Operating System.
- 7 Explain ls, cd commands with use, syntax and example.
- 8 Explain grep command with at least four possible attributes and examples.
